

Unit 4 Guided Notes — Geometry: Angles & Triangles

Name the relationship, write the equation, solve for x . Angles always add up to something.

 **By the end of this unit, I can...**

- Use angle relationships formed by parallel lines and a transversal
- Apply the Triangle Angle-Sum and Exterior-Angle Theorems
- Solve for unknown angles by writing and solving equations

Standard: NC.8.G.5

 **Blank Worksheet (PDF)**

 **Answer Key (PDF)**

 **Official PDF Packet**

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Name the relationship, write the equation, solve for x — angles always add up to something!

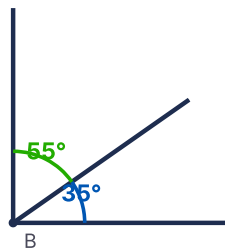
1 Angle Relationships

Fill the definitions and rules on your printed copy, then click to check.

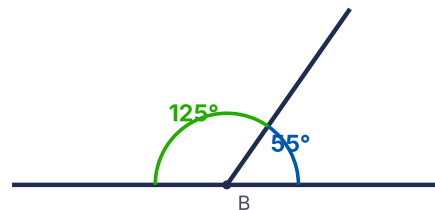
Relationship	Definition	Sum / Rule
Complementary	Two angles that add to °	
Supplementary	Two angles that add to °	
Vertical Angles		
Linear Pair		

Each angle is named with three letters — the vertex is the MIDDLE letter (e.g. $\angle ABD$ has vertex B). Arcs show which angle each measure belongs to.

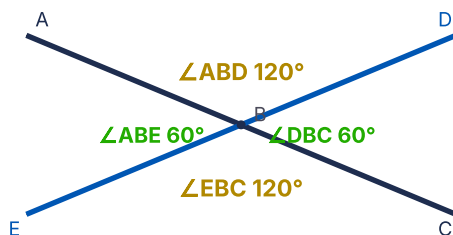
COMPLEMENTARY = 90°



SUPPLEMENTARY = 180°

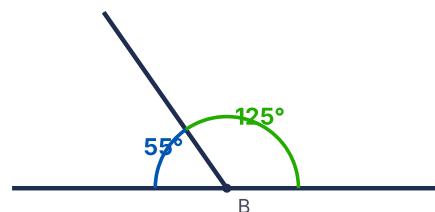


VERTICAL ANGLES



Opposite angles are EQUAL: $\angle ABD = \angle EBC$, $\angle DBC = \angle ABE$

LINEAR PAIR = 180°



Quick Check

MC Two angles form a linear pair. One measures 115° . What is the other?

✓ A linear pair adds to 180° . $180^\circ - 115^\circ = 65^\circ$.

Match the Relationship to its Rule

MATCH Click a relationship, then its rule.

Relationship

Complementary

Supplementary

Vertical Angles

Linear Pair

Rule

Always EQUAL

Add to 90°

Adjacent angles on a line $\rightarrow 180^\circ$

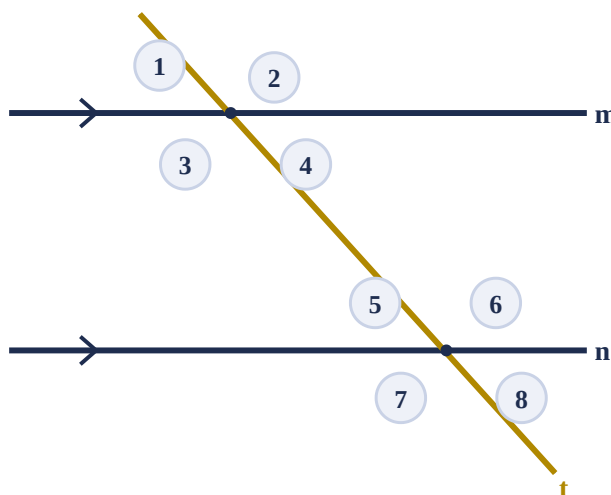
Add to 180°

0 of 4 matched

2 Parallel Lines & Transversals

REMEMBER: Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal). Same-Side Interior and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).

PARALLEL LINES & TRANSVERSAL



Two parallel lines ($m \parallel n$) cut by transversal t form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)

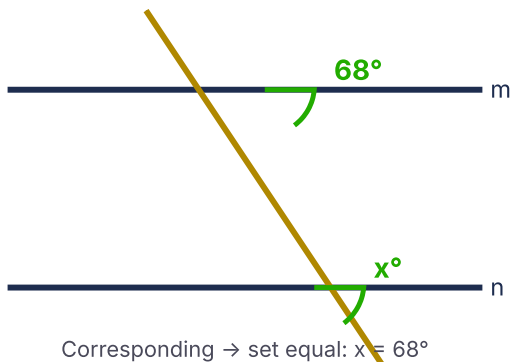
Fill the Relationship and Rule columns on your printed copy, then click to check.

Angle Pair Type	Relationship	Rule to Use	Example
Corresponding			$68^\circ \rightarrow x = 68^\circ$
Alternate Interior			$101^\circ \rightarrow x = 101^\circ$
Alternate Exterior			$142^\circ \rightarrow x = 142^\circ$
Vertical			$120^\circ \rightarrow x = 120^\circ$
Same-Side Interior			$134^\circ \rightarrow x = 46^\circ$

Same-Side Exterior

$$77^\circ \rightarrow x = 103^\circ$$

CORRESPONDING ANGLES ARE EQUAL



Worked Example — $\angle 1 = 3x + 15$ and $\angle 2 = 5x - 7$ (alternate interior angles)

Alternate interior angles are EQUAL: $3x + 15 = 5x - 7$

Subtract $3x$: $15 = 2x - 7$

Add 7: $22 = 2x$

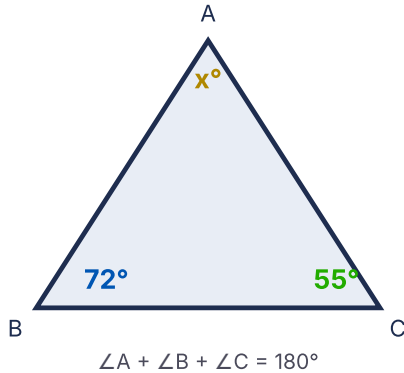
✓ **Divide by 2:** $x = 11 \rightarrow \text{angle} = 3(11) + 15 = 48^\circ$

3 Triangle Theorems

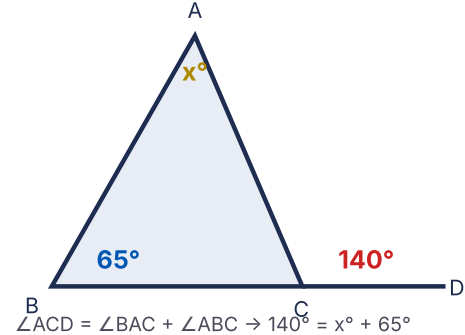
TRIANGLE ANGLE SUM: The three interior angles of a triangle always add up to 180° .

EXTERIOR ANGLE THEOREM: An exterior angle equals the SUM of the two non-adjacent (remote) interior angles.

TRIANGLE ANGLE SUM = 180°



EXTERIOR ANGLE THEOREM



Problem 1 — Triangle Angle Sum: $x + 72 + 55 = 180$

Solve: $x + 127 = 180 \rightarrow x = 53^\circ$

Problem 2 — Exterior Angle Theorem: $x + 65 = 140$

Solve: $x + 65 = 140 \rightarrow x = 75^\circ$

ERROR ANALYSIS Jordan's Exterior Angle Mistake — A triangle has interior angles 35° and 70° . Exterior angle = x . Jordan wrote $x = 70 - 35 = 35^\circ$.

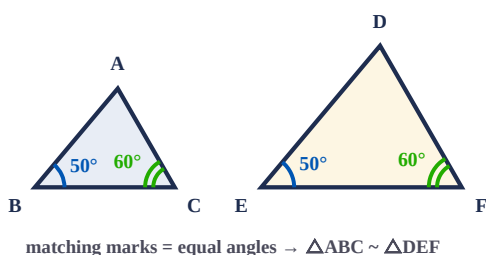
Jordan subtracted instead of adding. The exterior angle = SUM of the remote interior angles: $x = 35 + 70 = 105^\circ$.

4 Angle-Angle (AA) Similarity

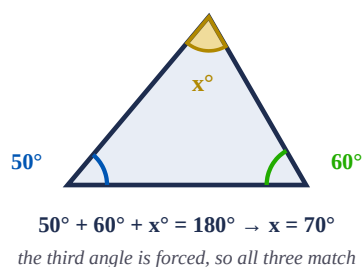
🔑 SIMILAR TRIANGLES: Two triangles are *similar* (written $\triangle ABC \sim \triangle DEF$) when they have the same shape but not necessarily the same size. Corresponding angles are equal and corresponding sides are in proportion.

🔑 ANGLE-ANGLE (AA) CRITERION: If two angles of one triangle are equal to two angles of another triangle, then the triangles are similar. You only ever need to check TWO pairs of angles.

ANGLE-ANGLE (AA) CRITERION

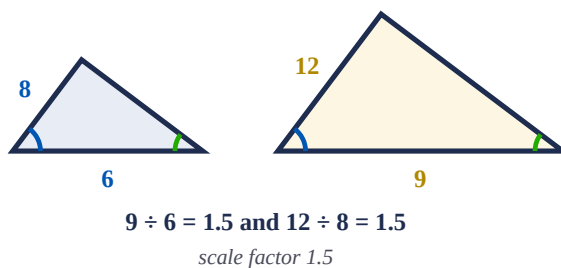


WHY TWO ANGLES ARE ENOUGH



🔑 WHY TWO ANGLES ARE ENOUGH: Every triangle's angles add to 180° . Once two pairs of angles match, the third pair has no choice but to match as well — so all three pairs are equal.

SIMILAR $\rightarrow$ SIDES IN PROPORTION



 **WATCH OUT:** AA needs two pairs of angles that *match each other* — not simply two angles given in each triangle.

Problem 1 — Are they similar? $\angle A = 50^\circ$, $\angle B = 60^\circ$ and $\angle D = 50^\circ$, $\angle E = 60^\circ$

Decide: Two pairs of angles are equal ($50^\circ = 50^\circ$ and $60^\circ = 60^\circ$), so by AA the triangles are similar: $\triangle ABC \sim \triangle DEF$.

Problem 2 — $\triangle ABC \sim \triangle DEF$ with $\angle A = 35^\circ$ and $\angle B = 85^\circ$. Find $\angle F$.

Solve: $\angle C = 180 - 35 - 85 = 60^\circ$. $\angle F$ corresponds to $\angle C$, so $\angle F = 60^\circ$.

Problem 3 — Similar triangles: sides 6 and 8 match sides 9 and x . Find x .

Solve: Scale factor $= 9 \div 6 = 1.5$, so $x = 8 \times 1.5 = 12$.

ERROR ANALYSIS Riley's AA Mistake — Triangle 1 has angles 40° and 75° . Triangle 2 has angles 40° and 70° . Riley wrote

“Each triangle shows two angles, so they are similar by AA.”

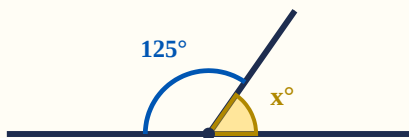
AA needs two pairs of angles that match *each other*. Only the 40° angles match here. Triangle 1's angles are $40^\circ, 75^\circ, 65^\circ$; Triangle 2's are $40^\circ, 70^\circ, 70^\circ$. The two sets are different, so the triangles are **not** similar.

Part 2 — Practice Problems

 **Identify the relationship, write the equation, and solve for x .** Type just the number, e.g. 55.

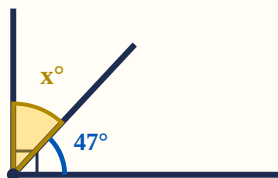
A. Angle Relationships — Solve for x

1) Supplementary Angles, given 125° .
 $x = ?$



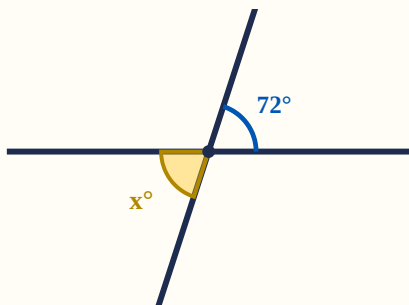
$x =$

2) Complementary Angles, given 47° .
 $x = ?$



$x =$

3) Vertical Angles, given 72° . $x = ?$



$x =$

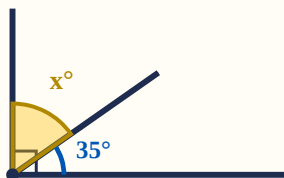
4) Linear Pair, given 138° . $x = ?$



$x =$

5) Complementary Angles, given 35° .

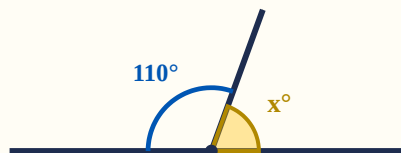
$x = ?$



$x =$

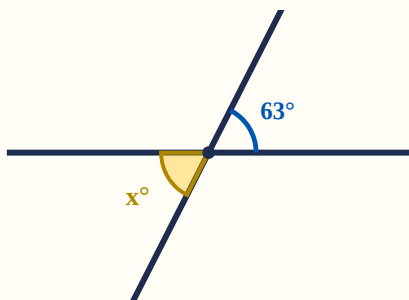
6) Supplementary Angles, given 110° .

$x = ?$



$x =$

7) Vertical Angles, given 63° . $x = ?$



$x =$

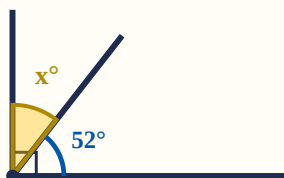
8) Linear Pair, given 157° . $x = ?$



$x =$

9) Complementary Angles, given 52° .

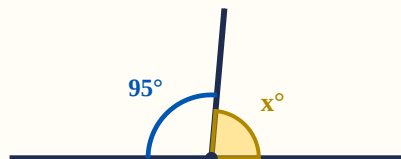
$x = ?$



$x =$

10) Supplementary Angles, given 95° .

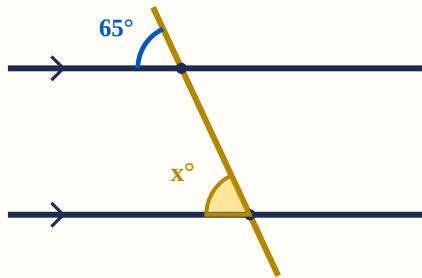
$x = ?$



$x =$

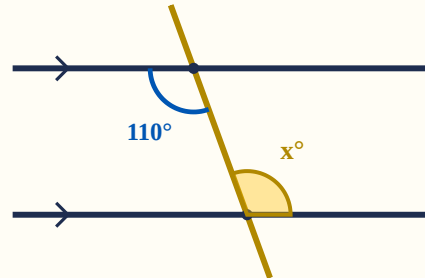
B. Parallel Lines & Transversals

1) Corresponding, given 65° . $x = ?$



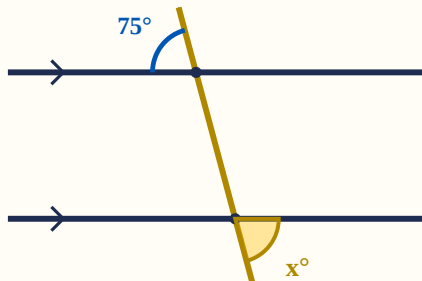
$x =$

2) Alternate Interior, given 110° . $x = ?$



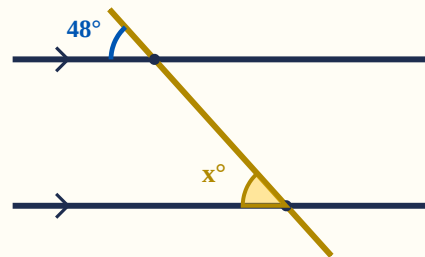
$x =$

3) Alternate Exterior, given 75° . $x = ?$



$x =$

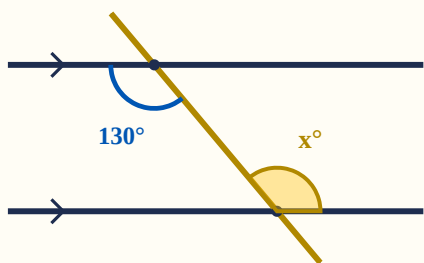
4) Corresponding, given 48° . $x = ?$



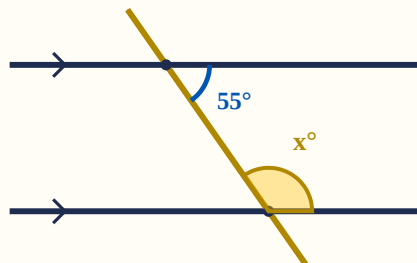
$x =$

5) Alternate Interior, given 130° . $x = ?$

6) Co-Interior, given 55° . $x = ?$

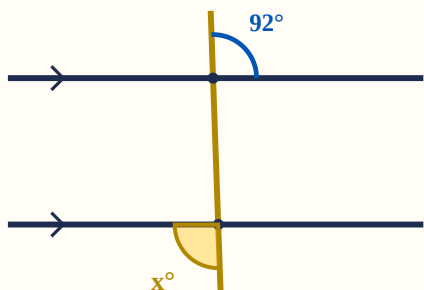


$x =$



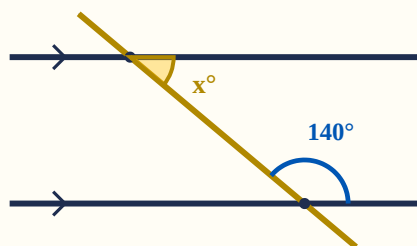
$x =$

7) Alternate Exterior, given 92° . $x = ?$



$x =$

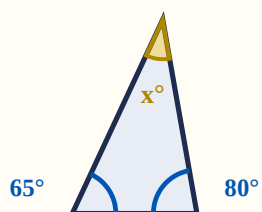
8) Co-Interior, given 140° . $x = ?$



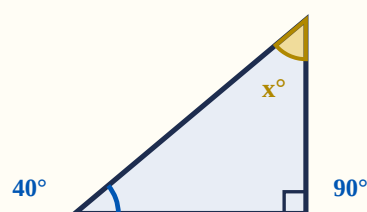
$x =$

C. Triangle Theorems

1) Angle Sum, given 65° , 80° . $x = ?$



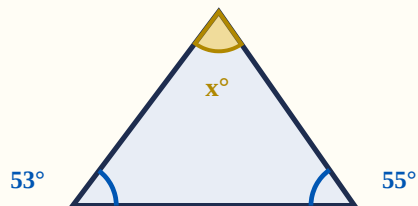
2) Angle Sum, given 40° , 90° . $x = ?$



$x =$

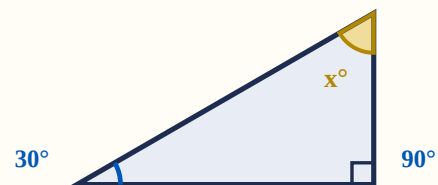
$x =$

3) Angle Sum, given 53° , 55° . $x = ?$



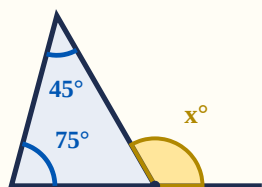
$x =$

4) Angle Sum, given 30° , 90° . $x = ?$



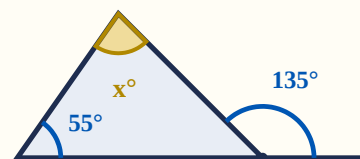
$x =$

5) Exterior Angle Theorem, remotes 45° , 75° . Exterior $x = ?$



$x =$

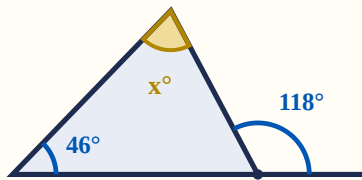
6) Exterior Angle Theorem, remote 55° , exterior = 135° . $x = ?$



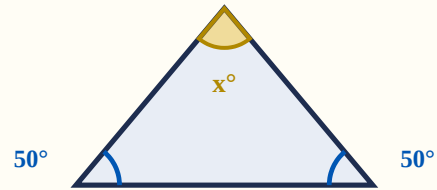
$x =$

7) Exterior Angle Theorem, remote 46° , exterior = 118° . $x = ?$

8) Angle Sum, given 50° , 50° . $x = ?$



$x =$



$x =$

D. Angle-Angle (AA) Similarity

1) AA Similarity — two angles match:
 50° and 60° . $x = ?$



Triangle 1

Triangle 2

$x =$

2) $\triangle ABC \sim \triangle DEF$, given 35° and 85° . $x = ?$



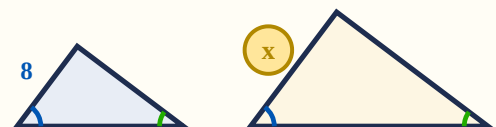
Triangle 1

Triangle 2

$x =$

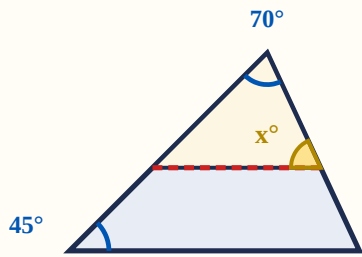
3) Parallel line inside a triangle, given
 70° and 45° . $x = ?$

4) Similar triangles — sides 6 and 8
 match 9 and x . $x = ?$



Triangle 1

Triangle 2

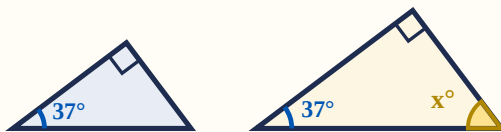


the dashed line is parallel to the base

$x =$

$x =$

5) Similar right triangles, given 37° . $x = ?$

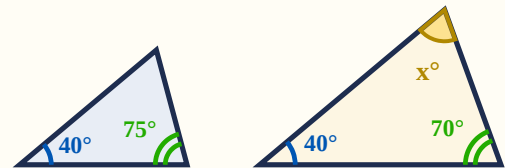


Triangle 1

Triangle 2

$x =$

6) Triangle 1: $40^\circ, 75^\circ$. Triangle 2: $40^\circ, 70^\circ$. $x = ?$



Triangle 1

Triangle 2

$x =$

← Back to Unit 4